

CHAOYI ZHANG

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EDUCATION

Ph.D. Zhejiang University 2022

School of Brain Science and Brain Medicine

Neurobiology; Mentor: Hailan Hu

B.S. Wenzhou Medical University 2016

School of Pharmacy

Biopharmaceutics

RESEARCH EXPERIENCE

Post Doc 2023-Current

PI: Guoping Feng

Massachusetts Institute of Technology, Cambridge, U.S.

- Developed a chronic functional ultrasound imaging (fUSI) platform for awake, head-fixed marmosets, achieving $15 \times 15 \times 15 \text{ mm}^3$ brain-wide coverage during task performance through a TPX-film cranial window
- Constructed a marmoset whole-brain vascular reference map and an NMI-based automated registration pipeline onto the MBM+SAM atlas; extended it to partial-brain fields of view using deep learning and to the mouse brain
- Applying the platform to brain-wide network dysfunction in SHANK3 mutant marmosets, with cross-species comparison to SHANK3 mouse models

Post Doc 2022-2023

PI: Gordon Fishell

Harvard Medical School, Boston, U.S.

- Investigated how wS1 circuits assemble to control whisking behavior during the development

Ph.D.

2016-2022

PI: Hailan Hu

Zhejiang University, Hangzhou, P.R. China.

- Set up the lab's freely moving cellular-resolution one-photon calcium imaging system
- Explored the roles and dynamics of different mPFC cell types in social competition behavior via in vivo optrode recording and miniature two-photon microscope

Summer Scholar

2016

PI: Minmin Luo

National Institute of Biological Sciences (NIBS), Beijing, P.R. China

- Explored the roles of dopamine neurons and serotonin neurons in reward behavior via fiber photometry recording and optogenetics

Undergraduate Research Assistant

2013-2016

PI: Hongyu Zhang, Xiaokun Li

- Validated the repair function of the basic fibroblast growth factor (bFGF) on the blood brain barrier after brain injury on mice model
- Led a team in “Challenge Cup” National Undergraduate Extracurricular Academic Science and Technology Works Competition

PUBLICATIONS

Zhang, C.*, H. Zhu*, Z. Ni*, Q. Xin, T. Zhou, R. Wu, G. Gao, Z. Gao, H. Ma, H. Li, M. He, J. Zhang, H. Cheng and H. Hu (2022). "Dynamics of a disinhibitory prefrontal microcircuit in controlling social competition." *Neuron*. 110(3): 516–531. **(Cover story, Co-first author)**

Fan Z, Chang J, Liang Y, Zhu H, **Zhang C**, Zheng D, Wang J, Xu Y, Li Q, Hu H*.(2023). Neural mechanism underlying depressive-like state associated with social status loss. *Cell*. 186: 1–17

Zhang, C., et al. Volumetric functional ultrasound in behaving marmosets: from chronic acoustic window to brain-wide imaging. *Manuscript in preparation*.

SKILLS

Experimental skills

Functional ultrasound imaging: volumetric and 4D row-column-addressed fUSI

Primate experimentation: marmoset chronic cranial-window and head-post surgery; head-fixation training; MonkeyLogic behavioral control

Optical and electrophysiological recording: two-photon imaging; single-cell calcium imaging via GRIN lens; multi-channel fiber photometry; in vivo optrode recording

Circuit manipulation: optogenetics and chemogenetics; focused ultrasound/sonogenetics

Histology and behavior: immunostaining and Western blot; social and cognition behavior assays

Computational and engineering skills

Languages: MATLAB, Python

Registration and atlases: AFNI, ANTs and FireANTs; NMI-based automated registration; marmoset MBM+SAM atlas; vascular reference-map construction

Analysis: large-scale imaging analysis; machine-learning decoding; DeepLabCut / DLC-Live; High-performance computing (Slurm)

Instrumentation: PPG biosensors; TTL synchronization; CAD and 3D printing

Imaging and design: ImageJ; Adobe Illustrator; 3D Slicer; Autodesk Fusion

TALKS AND POSTERS

- **Gordon Research Conference on In Vivo Ultrasound Imaging** 2026

Talk in GRS and Poster in GRC

- **Ray Wu Symposium, Westlake University** 2023

Oral presentation; Hangzhou, China

- **NBO Webinar: Reading the brain activity via calcium imaging** 2022

Invited lecture in NBO (Nanjing Brain Observatory), Online

- **Nanjing international symposium on “Brain–Mind–Intelligence” (2021)** 2021

Oral presentation; Online

- **“Brain and Psychology” National Doctoral Forum** 2020

Oral presentation (Outstanding lecture award); Hangzhou, China

TEACHING EXPERIENCE

Teaching Assistant 2020

General Education Course: Exercise and Brain

Zhejiang University, Hangzhou, China

- Organized and led tutorials on topics about neuroscience and sports

Guest Lecturer and Consultant 2019

National single-cell resolution *in vivo* calcium imaging technology workshop

Hangzhou, China

- lecture title: Reading the brain activity via calcium imaging
- Surgery and freely moving single-cell resolution calcium imaging demonstration

AWARDS, SCHOLARSHIPS, HONORS

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| ● J. Douglas Tan Postdoctoral Fellowship, MIT | 2024-2026 |
| ● Excellent Doctoral Dissertations of Zhejiang University | 2023 |
| ● Ray Wu Prize | 2022 |
| ● Excellent graduate of Zhejiang Province | 2022 |
| ● Dean's Scholarship | 2021 |
| ● Funding for Excellent Doctoral Dissertations, Zhejiang University | 2021 |
| ● Championship of Zhejiang Province College Tennis Tournament | 2020 |
| ● National scholarship | 2020 |
| ● Student Leadership Award, Zhejiang University | 2019 |
| ● Excellent graduate of Zhejiang Province | 2016 |

CONTRIBUTIONS TO SCIENCE

Brain-wide functional ultrasound imaging in behaving marmosets (2023-present, Dr. Guoping Feng lab, MIT)

Existing methods for recording neural activity in primates trade brain-wide coverage against spatial resolution, which limits what can be learned about how distributed circuits coordinate to produce behavior. Functional

ultrasound imaging offers a way out, and I have spent my postdoc making it work in the awake marmoset. I developed a chronic fUSI approach for head-fixed animals achieving $15 \times 15 \times 15 \text{ mm}^3$ of brain coverage during task performance through a TPX-film cranial window, implemented on both a motorized linear-array scanner and a row-column-addressed 4D probe, and built the reconstruction and analysis chain behind it.

Because a single imaging session is of limited value until it can be compared with every other one, a large part of this work has gone into registration. I constructed a whole-brain marmoset vascular reference map and an automated pipeline that aligns it to an existing marmoset MRI atlas, and have since extended the approach to partial-brain fields of view using deep learning — the previously published vascular brain-positioning approach required whole-brain coverage — and to the mouse brain. I intend to release both the marmoset and mouse reference maps publicly. Using this framework I identified distinct regions activated by cues versus reward or air-puff stimuli in an auditory cue-outcome association task and introduced a live social interaction paradigm whose activation pattern differs from the social video-watching tasks conventional in non-human primate fMRI (manuscript in preparation). In parallel I built MarmoFACT, a synchronized facial-action and cardiac monitoring pipeline that gives autonomic and expressive readouts time-locked to imaging. Together these are aimed at brain-wide network dysfunction in SHANK3 mutant marmosets and, longer term, at circuit-level biomarkers for autism spectrum disorder.

The role of mPFC in social competition behaviors (2016-2022, Dr. Hailan Hu lab, Zhejiang University)

The health and quality of life of animals strongly depend on their dominant status, which is acquired through repeated bouts of social competition. Our lab's previous studies have confirmed the key role of the medial prefrontal cortex (mPFC) in controlling how the mice performed in social competition (*Wang et al. 2011, Science*; *Zhou et al. 2017, Science*). However, it has remained unexplored how different mPFC cell types engage in social competition. Approximately 80% of cortical neurons are excitatory pyramidal (PYR) neurons, which are generally responsible for the processing and transmitting information. However, the remaining 20% are diverse inhibitory interneurons (such as PV+, SST+ and VIP+ neurons), form microcircuits with pyramidal neurons, also highly involved in the precise control of information processing within the brain. Over six years I used a multidisciplinary approach, combining optogenetics, chemogenetics, fiber photometry, cutting-edge miniature two-photon endoscope calcium imaging and *in vivo* optrodes recording to elucidate the organization and function of mPFC underlying social competition behaviors.

During the year, I identified the VIP-PV-PYR disinhibitory motif forming the core of a dynamic dmPFC microcircuit to control social competition. My study revealed that VIP neurons, a subset of interneurons that targets other inhibitory cells, play “VIP” roles in mPFC microcircuits. VIP neurons initiated sequential activities, followed by PYR and PV neurons in winning behaviors and they regulated PYR neurons activity through “pull-push” mechanism. In addition, I found the unappreciated role of PV neurons in mediating VIP neuronal disinhibition in mPFC microcircuits. My study further elucidated how mPFC processes social information and regulates social behavior (*Zhang et al. 2022 Neuron; Cover story*).